FLOW THROUGH PIPES

- A pipe is a close Conduit used for Carrying fluids under pressure.
- The flowing fluid is subjected to resistance due to
  (i) Shear forces between fluid particles and boundary walls of pipe.
  (ii) between fluid particles themselves due to viscosity of fluid.
- Loss of energy of the flowing fluid in the direction of flow to over come resistance and depends upon the type of flow
  (i) laminar or (ii) Turbulent.

- Round pipes are most frequently used to carry fluids.
  eg: Water for water distribution network
  oil for oil transportation.
  air for pneumatic system.
  gas and steam for commercial and domestic requirement
- pressure in pipes may be above or below atmospheric. The negative pressure is restricted to Saturation Vapour pressure at that temperature otherwise liquid will evaporate and Constitute liquid vapour mixture and may result in obstruction to flow. Besides this, Cavitation may occur which is detrimental to the system.

Note: Cavitation phenomenon

When pressure falls below the evaporation pressure, liquid boils and large number of bubbles of vapour are formed. These bubbles are Carried by Stream to higher pressure zones, where vapour Condenses and bubles Suddenly collapse resulting in the formation of Cavity and surrounding liquid rushes to fill it. This results in giving rise to a very high local pressure.

Formation of Cavity and high pressure are repeated many thousand times a second. This Causes pitting of metallic surface and metal fails due to fatigue and corrosion.

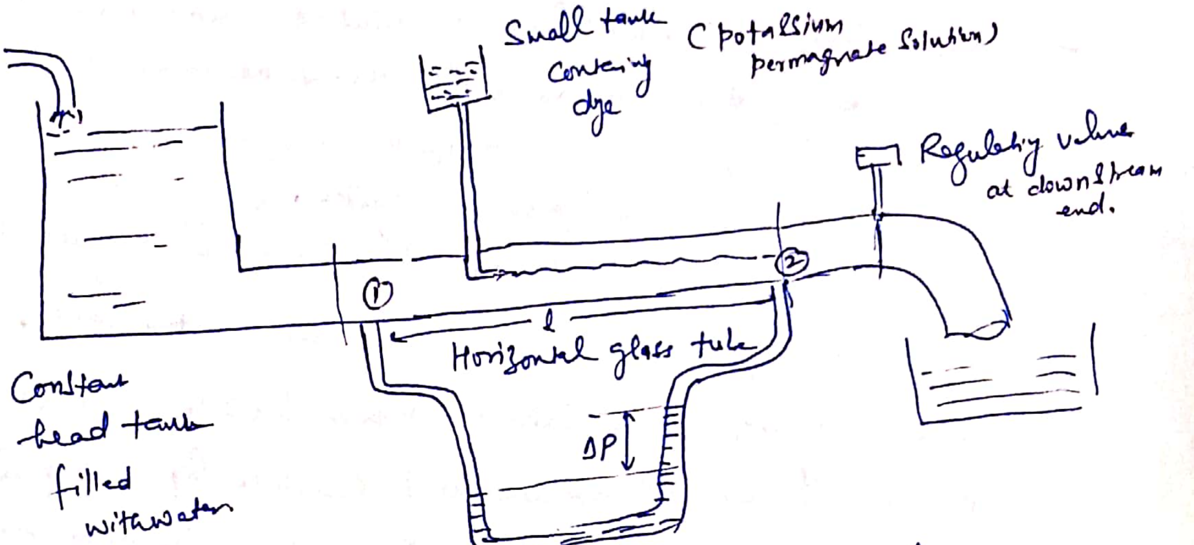
– Hence pipes should convey fluids at a pressure greater than vapour pressure.

② - When a pipe discharges fluid to atmosphere, the pressure at exit must be atmospheric. Similarly fluid entering from ambient must be at atm. pressure.

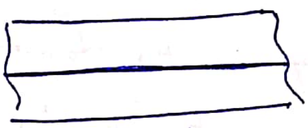
- flow in a confined passage is fully developed if the velocity profile does not alter along the flow.

- Fully developed flow can occur only in constant area symmetrical passages of pipes, annuli and parallel plates.
- Fully developed flow can be laminar or turbulent depending on Reynolds number.
- In fully developed pipe flow pressure drops linearly along the length of pipe-line i.e. pressure gradient along the flow remains constant.
- Average velocity of flow is taken as base velocity for assumption of uniform velocity at each cross section.

Reynold's experiment.



- velocity of flow was adjusted by regulating valve.
- liquid dye (eg potassium permagnet) was introduced into the flow.

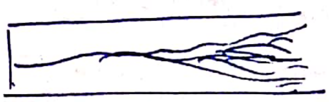


laminar flow

dye remain in straight path at low velocity

Transition

with increase of velocity at critical value the dye showed irregularity in straight path and flow waviness



turbulent flow

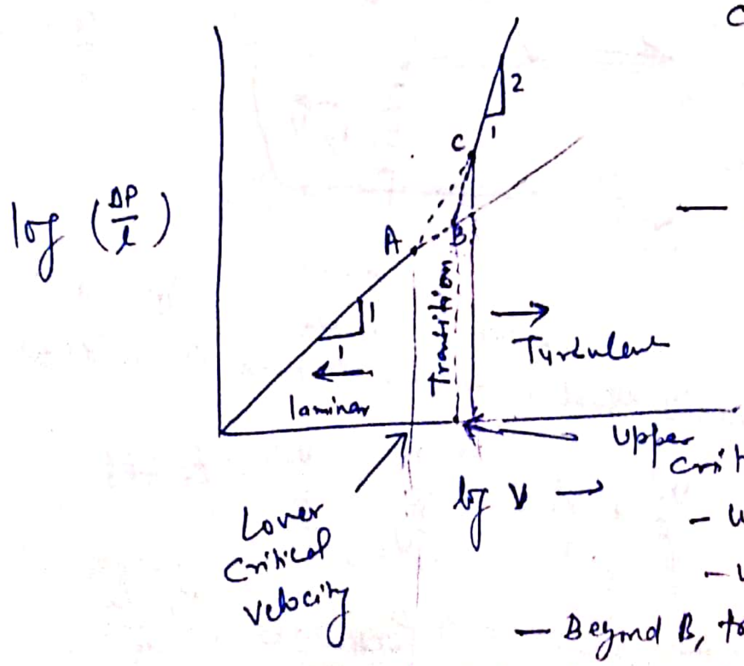
with further increase of velocity the dye became more intense and diffused over entire area of cross-section. This is the process of changing laminar to turbulent flow.

- Reynold's deduced that
- at low velocity the fluid particles moves in parallel layer or laminaes and the regime is called laminar flow.
- At higher velocity the flow was turbulent.
- Reynolds concluded that occurrence of laminar and turbulent flow is governed by relative magnitude of inertia and viscous forces. because for the fluids having very small viscosity the viscous forces become predominant at low velocity and at higher velocity the inertia forces are predominant.

$$Re_d = \text{Reynolds number} = \frac{\text{Inertia force}}{\text{viscous force}} = \frac{\rho V d}{\mu}$$

In order to get the local change of nature of flow

In order to get differential pressure manometer between two sections $\underline{1}$ and $\underline{2}$ located l apart can be used to get pressure drop ΔP as a function of velocity.



- Point B = upper critical point corresponding velocity for particular d, ρ, μ is called upper critical velocity.
- Point A = Lower critical point corresponding velocity for particular d, ρ, μ is called lower critical velocity.

- upto point A, $h_f \propto V$
- upto point B, $h_f \propto V$
- Beyond B, transition occurs upto point C.
- Beyond C, curve is straight line with slope 1.72 – 2.00.
- If velocity is gradually reduced line, BC will not be traced but CA will be traced.

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- for small values of v the plot is straight line and slope is equal to unity.

<u>Upper critical Reymolds number</u>

- ^upto increasing the velocity certain critical value, the lommar flow may be maintained and called upper critical reynolds number. (between 2700 - 3000)

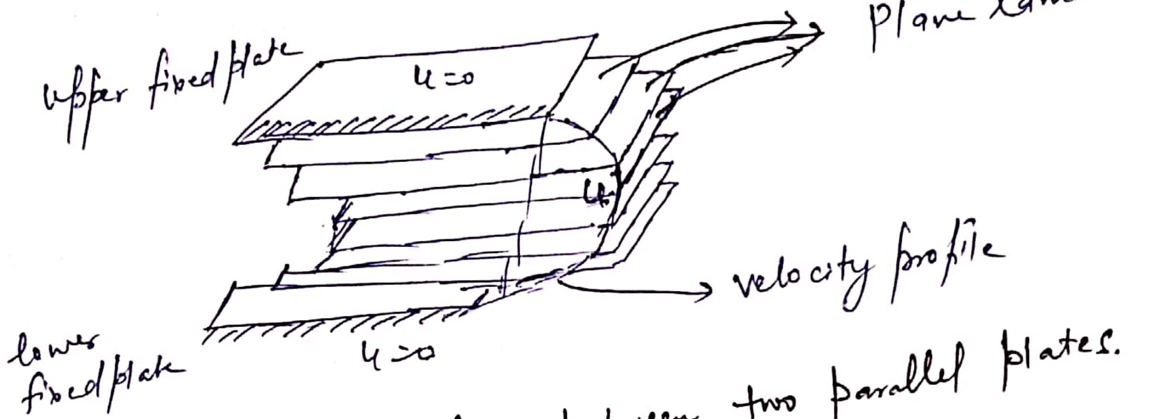
<u>Lower critical Reynolds number</u>

- If the velocity is gradually reduced fow a higher value to change the nature of flow from turbulent to laminar flow. The critical velocity to which the flow must be reduced is called lower critical velocity. (L.C.Reynolds number = 2000)

- Transition between (2000 - 3000).

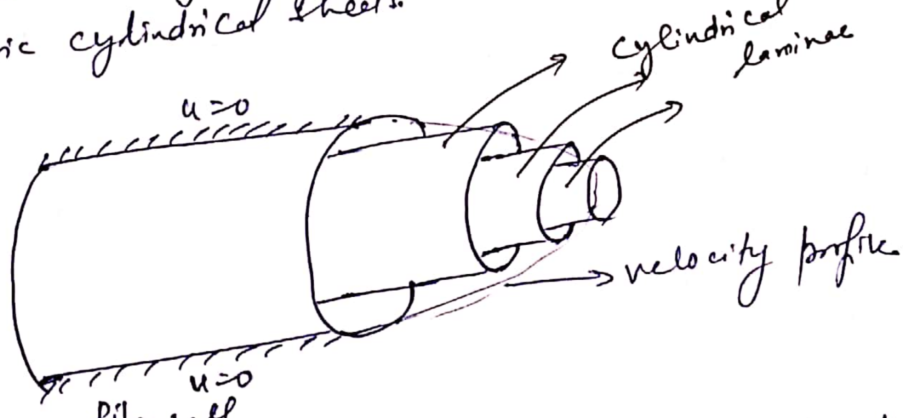
Laminar and turbulent flows

- Flow is said to be laminar if the flow is characterised by fluid particles flowing in laminae i.e. thin sheets.
- Shape of laminae depends on shape of boundaries of the passage.
- For flow between parallel plates, laminae are plane sheets parallel to each other.



laminar flow between two parallel plates.

- For flow through a round pipe, laminae are thin concentric cylindrical sheets.



- In turbulent flow fluid particles tends to mix in diffsent laminae.
- flow is likely to be lamnar if 1) velocity (V) is low.
  2) diameter (d) is small
  3) density (ρ) is low
  4) viscosity (μ) is high

these parameters grouped in a dimensionless no. (Reynolds No.) $= Re = \frac{\rho V d}{\mu}$

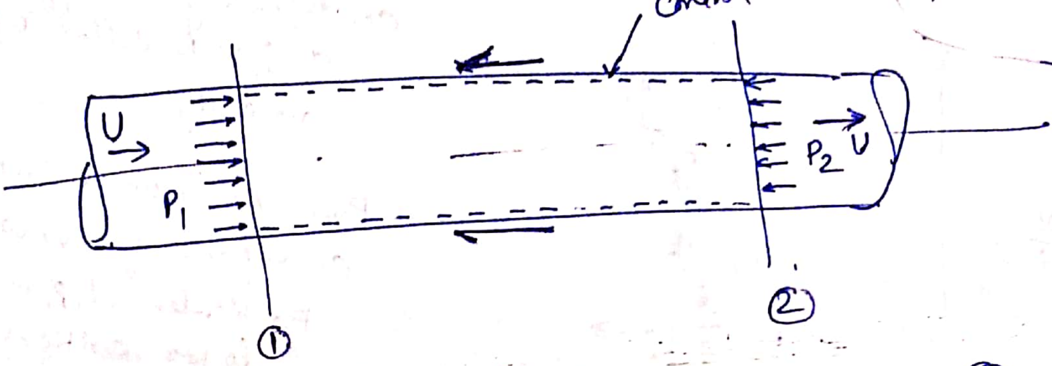
Friction loss in pipe flow : Darcy-weisback equation

- Consider steady flow through a pipe of constant diameter. The average velocity at each cross section remains the same.

from continuity equation

$$U = Q/A$$

- Static pressure p drops along the direction of flow because stagnation pressure drops due to loss of energy in overcoming friction as flow occurs.



Applying Bernoulli's equation between sections 1 and 2.

$$\frac{P_1}{\rho g} + \cancel{\frac{V_1^2}{2g}} + \cancel{z_1} = \frac{P_2}{\rho g} + \cancel{\frac{V_2^2}{2g}} + \cancel{z_2} + h_f$$

h_f = loss of energy (head) due to friction $= \dfrac{P_1 - P_2}{\rho g}$

= Difference in pressure heads between two sections.

Consider a fluid control volume between section 1 and 2

Pressure force along the flow

$$= (P_1 - P_2)\frac{\pi}{4}d^2 \quad \text{—(i)}$$

Frictional force due to shearing at the pipe wall resisting the flow.

$$= C_D \cdot \frac{1}{2}\rho V^2 . \pi d l \quad \text{—(ii)}$$

equating (i) and (ii) for equilibrium.

$$(P_1 - P_2)\frac{\pi}{4}d^2 = C_D \cdot \frac{1}{2}\rho V^2 . \pi d l$$

$$\frac{P_1 - P_2}{\rho g} = \frac{4C_D \cdot l V^2}{2gd}$$

Note:

C_D = Coefficient of drag

$$\boxed{C_D = \frac{\text{Drag force }(D)}{\frac{1}{2}\rho U_\infty^2 A}}$$

$$\boxed{h_f = \frac{f l V^2}{2gd}}$$

$f = 4C_D$ = friction factor for pipe roughness.

Darcy-weisbach equation used for computing loss of head due to friction in pipe. Also called major losses.

It has been shown that for laminar flow $f = \frac{64}{Re}$ for $Re < 2000$

for turbulent flow

$f = \frac{0.316}{Re^{1/4}}$ for $4000 < Re < 10^5$

Experimentally it has been found that

$$\boxed{f = fn(\text{Reynolds no., relative Roughness of pipe surface})}$$

These are classified as major losses because the head loss due to friction is much more than the loss of energy incurred by other causes.

- If Q = discharge through a pipe $= \frac{\pi}{4}d^2 \times V$

$$\Rightarrow V = \frac{4Q}{\pi d^2}, \quad h_f = \frac{fl}{2gd} \cdot \frac{16Q^2}{\pi^2 d^4}$$

$$\boxed{h_f = \frac{flQ^2}{12d^5}}$$

Note:

- friction factor decreases with increase in Reynolds number.
- Relationship between friction factor, Reynolds number and relative roughness is graphically represented by Moody's diagram.
- These plots are based on extensive experimental work of Nikuradse on artificially roughened pipes.

Tutorial problems (Solved)

Que: Water flows in a 2 cm diameter pipe. Taking kinematic viscosity of water as 0.0098 stoke, calculate the largest discharge for which the flow will be laminar.

Solution For flow to be in laminar character

$$Re < 2000$$

$$Re = \frac{\rho V d}{\mu} = \frac{Vd}{\nu} = \frac{V \times 2\times10^{-2}}{0.0098\times10^{-4}} < 2000$$

$$V < 0.098 \text{ m/s}$$

$$Q = A \times V = \frac{\pi}{4} d^2 \times 0.098 = \frac{\pi}{4} \times (.02)^2 \times 0.098$$

$$= 3.078 \times 10^{-5} \text{ m}^3/\text{s}$$

$$= .03078 \text{ Litre/s}$$

$$= 1.847 \text{ L/min.}$$ (largest discharge for flow to be laminar.)

Tutorial Problem (Solved)

Que An oil of mass density 950 kg/m³ and dynamic viscosity of 1.0 NS/m² is carried at the rate of 0.14 m³/s through a 30 cm diameter pipe over 1000 m length. Due to increase in environmental temperature, the viscosity of oil changes by a factor of 8. If the same quantity of oil is to be conveyed compare the cost of pumping.

Solution **Case I:**

$$Q = 0.14 \text{ m}^3/\text{s}$$

$$= A \times V = \frac{\pi}{4} \times (0.3)^2 \times V$$

$$\Rightarrow V = 1.98 \text{ m/s}$$

$$Re = \frac{\rho V d}{\mu}$$

$$= \frac{950 \times 1.98 \times 0.3}{1}$$

$$= 564.3 < 2000$$ (laminar regime)

(11)

friction factor $= \frac{64}{Re}$

$= \frac{64}{564.3} = .1134$

head loss due to friction $=$ $h_f = \frac{f l v^2}{2gd} = \frac{.1134 \times 1000 \times (1.98)^2}{2 \times 9.81 \times .3}$

$= 75.54$ m

Power required for pumping $= \rho Q g h$

$= 950 \times 0.14 \times 9.81 \times 75.54$

$= 98560 W = 98.56$ kW

Case 2 :

The change in viscosity due to increase in environmental temperature

$\mu_2 = \frac{\mu_1}{8} = \frac{1}{8}$ (viscosity of liquid decreases with increase in temperature)

$Re = \frac{\rho V d}{\mu} = \frac{950 \times 1.98 \times 0.3}{1/8}$

$= 4514.4 > 3000$ (turbulent flow)

$f = \frac{0.316}{(Re)^{0.25}} = .03855$

$h_f = \frac{f l v^2}{2gd} = \frac{.03855 \times 1000 \times (1.98)^2}{2 \times 9.81 \times 0.3}$

$= 25.67$ m

Now, Power required $= 950 \times 0.14 \times 9.81 \times 25.67$

$= 33501 W = 33.5$ kW

Ratio of pumping cost $= \frac{33.5}{98.56} = 0.34$

Tutorial problem (Solved)

Problem water is supplied to a town for 3 lakhs residents from a reservoir 5 km away from the town. The per capita consumption of water per day is 150 litres and half of the daily supply is pumped in 8 hours. The heighest and lowest water levels in the reservoir are 150 m and 100 m, respectively and the delivery end of main supply pipe is at 25 m above the reference level. If the head required at the delivery end is 15 m. Make calculations for the diameter of supply main. Take friction factor $f = 0.04$ in the Darcy's equation.

Solution

Total water supply required (per day)

$$= 3,00,000 \times 150 \times 10^{-3} \; m^3$$

$$= 45000 \; m^3$$

half of daily supply is pumped in 8 hours

$$q = \frac{45000 \; m^3}{2 \times 8 \times 3600 \; s} = 0.78125 \; m^3/s$$

Case I: water level in the reservoir is at highest level.

Head available $= 150 - 25 - 15 = 110$ m

$$h_f = \frac{flv^2}{2gd} = \frac{flQ^2}{12 d^5}$$

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$$110 = \frac{.04 \times 5000 \times (0.78125)^2}{12\ d^5}$$

$$d = .62\ m = 62\ cm$$

Case II: Water level in the reservoir is at the lowest level.

Head available

$= 100 - 25 - 15$

$= 60m$

$$h_f = 60 = \frac{f\ l\ Q^2}{12 d^5}$$

$$60 = \frac{0.04 \times 5000 \times (0.78125)^2}{12 \times d^5}$$

$$d = 0.70\ m = 70\ cm$$

- Now if we choose 62 cm diameter pipe then it will not be possible to get required discharge at required head.
- If we chose 70 cm diameter then available head will be more than sufficient to supply required quantity of discharge at required head. Hence required quantity can be made available by regulating quantity using a value.

Minor losses

- The head loss due to friction in a pipe is known as major energy loss.

- Loss of energy on account of change in velocity of flowing fluid (either in magnitude or direction) is called minor energy loss.

- In long pipes these are small as compared to major losses due to friction but in short length pipes these losses may be of considerable amount.

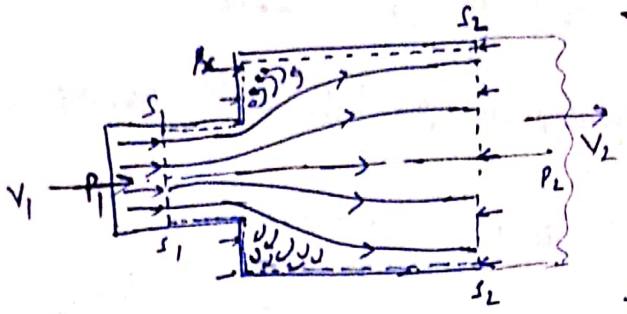
- Change in magnitude of velocity may be due to change in cross sectional area of the flow passage

  eg. Sudden enlargement
  Sudden contraction

- Change in direction of velocity is due to change in direction of flow passage.

  eg. pipe bends.

(1) loss of energy due to sudden enlargement



— Due to sudden charge of pipe dia. the flow from smaller pipe is not able to follow the abrupt change of boundary and flow separates from the boundary and consequent eddying motion in the corner region. and a loss of head occurs. loss of energy is ultimately dissipated as heat.

Loss of head can be calculated by taking a fluid control volume $S_1 S_1 S_2 S_2$

by Continuity equation $\quad A_1 U_1 = A_2 U_2 = Q$

by momentum Conservation $\rightarrow P_1 A_1 + P_x (A_2 - A_1) - P_2 A_2 = \rho Q (U_2 - U_1)$

Force acting on control volume in direction of flow.

P_x = mean pressure of liquid eddies over area $A_2 - A_1$.

$$\boxed{\begin{aligned} \text{Force} &= \text{Rate of change of momentum} \\ &= \dot{m}(V_2 - V_1) \\ &= \\ \dot{m} &= \rho_1 A_1 U_1 = \rho_2 U_2 A_2 \\ &= \rho_1 Q_1 = \rho_2 Q_2 \\ &= \rho Q \end{aligned}}$$

Experimentally it has been found that pressure P_x in the stagnation zone just after expansion equals P_1

$$P_1 \cancel{A_1} + P_1 A_2 - P_1 \cancel{A_1} - P_2 A_2 = \rho Q (U_2 - U_1)$$

$$(P_1 - P_2) A_2 = \rho Q (U_2 - U_1)$$

$$\frac{P_1 - P_2}{\rho g} = \frac{(U_2 - U_1) U_2 \cancel{A_2}}{g \cancel{A_2}} = \frac{U_2 (U_2 - U_1)}{g} \longrightarrow (A)$$

Applying Bernoulli's equation.

$$\frac{P_1}{\rho g} + \frac{U_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{U_2^2}{2g} + Z_2 + h_e$$

h_e ↓ loss of head due to sudden enlargement

$$h_e = \frac{P_1 - P_2}{\rho g} + \frac{U_1^2 - U_2^2}{2g} \qquad \text{from (A)}$$

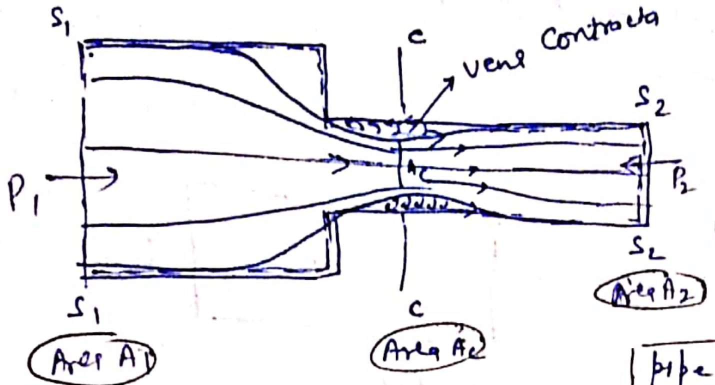
$$= \frac{2U_2^2 - 2U_1 U_2 + U_1^2 - U_2^2}{2g}$$

$$\boxed{h_e = \frac{U_1^2 + U_2^2 - 2U_1 U_2}{2g} = \frac{(U_1 - U_2)^2}{2g}}$$

$$\boxed{V_2 = \frac{A_1 V_1}{A_2}}$$

$$\boxed{h_e = \left(1 - \frac{A_1}{A_2}\right)^2 \frac{V_1^2}{2g}}$$

(2) loss of head due to Sudden Contraction



(i) In the region just upstream of junction there is converging or accelerating flow and no considerable loss of energy occurs.

(ii) Immediately down stream of the junction, when the liquid flows through narrower pipe, a vena contracta is formed, after which stream of liquid again fills completely the narrower pipe.

- In between vena contracta and walls of pipe, alot of eddies are formed, which causes considerable dissipation of energy.

- Between vena contracta and section 2-2 at a distance away from it, the flow pattern is similar as was in sudden enlargement.

- The loss of head due to sudden contraction is actually due to sudden enlargement from vena contracta to smaller pipe.

$$h_c = \frac{(V_c - V_2)^2}{2g} = \frac{V_2^2 (V_c/V_2 - 1)^2}{2g}$$

from continuity equation $A_c V_c = A_2 V_2 \Rightarrow \frac{V_c}{V_2} = \frac{A_2}{A_c}$

$$h_c = \frac{V_2^2 \left(\frac{A2}{A_c} - 1\right)^2}{2g} = \frac{V_2^2}{2g}\left(\frac{A2}{A_c} - 1\right)^2$$

$$\boxed{C_c = \frac{A_c}{A_2}}$$

$$\boxed{h_c = \frac{V_2^2}{2g}\left(\frac{1}{C_c} - 1\right)^2}$$

Value of C_c depends upon $\frac{A_2}{A_1}$ or $\frac{d_2}{d_1}$,

Take $C_c = 0.62$

$$h_c = 0.375 \frac{V_2^2}{2g}$$

Generally $\boxed{h_c = \frac{0.5 V_2^2}{2g}}$

(3) Loss of head at the entrance to pipe from the large vessel

- When liquid enters in a pipe from a large vessel, a loss of energy occurs.
- The flow pattern at entrance of the pipe is similar to that in case of sudden contraction. Hence similar energy loss is taken
- For sharp edge entrance, this loss is slightly more than rounded or bell mouthed entrance.

For sharp edged entrance $\boxed{h_i = 0.5 \frac{V^2}{2g}}$ V = velocity of liquid in pipe.

(4) Loss of head at the exit of pipe

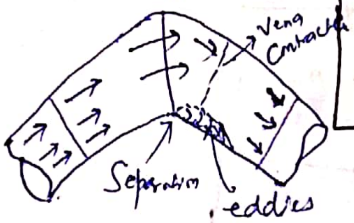
- The liquid at leaving end of pipe possesses some K.E. corresponding to velocity at outlet of pipe.
- The energy is dissipated either in form of free jet in case of free end or is lost in tank or reservoir in form of turbulence, diffusion, sound or heat

$$\boxed{h_e = \frac{V^2}{2g}}$$

V = velocity at outlet of pipe.

(5) Loss of head due to bend in pipe

- Due to bend, there is change in direction of flow which results in loss of energy. This is due to separation of flow from inner wall boundaries and consequent formation of eddies.



$$\boxed{h_b = K \frac{V^2}{2g}}$$

K depends on
- Angle of bend
- radius of curvature of bend
- Diameter of pipe.

(6) Loss of energy in various pipe fittings

- loss of energy due to valves, coupling etc.

$$h_e = K \frac{V^2}{2g}$$

V = mean velocity of flow in pipe

K = Coefficient depends upon the type of pipe fitting.

Tutorial Problem (Solved)

The diameter of a water pipe is suddenly enlarged from 350 mm to 700 mm. The rate of flow through it is 0.25 m^3/s and the pressure in the smaller pipe is 7.5 N/m^2. Calculate

(i) loss of head in the enlargement

(ii) Power loss due to enlargement, and

(iii) Pressure in the larger pipe if pipe is horizontal.

Solution

$$Q = .25 = A_1 V_1 = A_2 V_2$$

$$= \frac{\pi}{4} d_1^2 \times V_1 = \pi/4 \cdot d_2^2 \; V_2$$

$$.25 = \frac{\pi}{4} \times (.35)^2 \; V_1 = \pi/4 \times (.7)^2 V_2$$

$$\Rightarrow V_1 = 2.598 \approx 2.6 \, m/s$$

$$V_2 = .65 \, m/s$$

(i) loss of head due to sudden enlargement

$$h_e = \frac{(V_1 - V_2)^2}{2g} = \frac{(2.6 - .65)^2}{2g} = 0.1938 \text{ m head of water}$$

(ii) Power lost in the enlargement

$$= \rho q g h$$

$$= 1000 \times .25 \times 9.81 \times .1938 = 475.31 \text{ W}$$

(iii) Applying Bernoulli's eqn.

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2 + h_e$$

$$\frac{7.5}{1000 \times 9.81} + \frac{2.6^2 - .65^2}{2g} - .1938 = \frac{P_2}{\rho g}$$

$$7.645 \times 10^{-4} + .323 - .1938$$

$$\frac{P_2}{\rho g} = 7.645 \times 10^{-4} + .323 - .1938$$

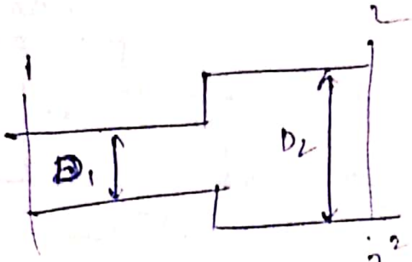
$$P_2 = .1299 \times 1000 \times 9.81$$

$$= 1275 \, N/m^2$$

(19) Tutorial problem (contd)

Que: A Horizontal pipe of diameter D_1 has a sudden expansion to a diameter D_2. For a given discharge at what ratio D_1/D_2 would the differential pressure on either side of the expansion be maximum? what is the corresponding loss of head and differential pressure head?

Solution



head loss due to enlargement, $h_e = \dfrac{(V_1-V_2)^2}{2g}$

Applying Bernoulli's eqn.

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho_2 g} + \frac{V_2^2}{2g} + z_2 + h_e$$

$$\frac{P_1-P_2}{\rho g} = \frac{V_1^2}{2g} - \frac{V_2^2}{2g} + \frac{(V_1-V_2)^2}{2g}$$

Applying Continuity eqn.

$$A_1 V_1 = A_2 V_2 \Rightarrow V_2 = \left(\frac{D_1}{D_2}\right)^2 V_1$$

$$\pi/4\, D_1^2\, V_1 = \pi/4\, D_2^2\, V_2$$

$$\boxed{V_2 = x^2 V_1}$$

Let $\boxed{x = D_1/D_2}$

$P_1 - P_2 = \Delta P$ = differential pressure

$$\left(\frac{\Delta P}{\rho g}\right) = \frac{V_1^2 - x^4 V_1^2 - (1-x^2)^2 V_1^2}{2g}$$

$$\left(\frac{\Delta P}{\rho g}\right) = \frac{V_1^2}{2g}\left[1 - x^4 - (1-x^2)^2\right]$$

for maximum or minimum $\dfrac{d(\Delta P/\rho g)}{dx} = 0 \Rightarrow \dfrac{V_1^2}{2g}\left[0 - 4x^3 - 2(1-x^2)(-2x)\right] = 0$

$$-4x^3 + 4x - 4x^3 = 0 \Rightarrow 2x^3 - x = 0$$

$$(2x^2 - 1)x = 0$$

for maxm. $\dfrac{d^2(\Delta P/\rho g)}{dx^2} = (-ve)$

$$\Rightarrow x = \frac{1}{\sqrt{2}} \Rightarrow \boxed{D_2 = \sqrt{2}\, D_1}$$

$$h_e = \frac{(V_1-V_2)^2}{2g} = \frac{V_1^2}{2g}(1-x^2)^2 = \frac{1}{4}\frac{V_1^2}{2g}$$

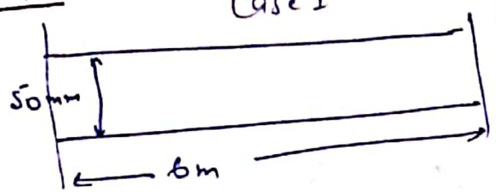
$$\Delta P/\rho g = \frac{1}{2}\frac{V_1^2}{2g} \leftarrow \frac{\Delta P}{\rho g} = \frac{V_1^2}{2g}\left[1 - \frac{1}{4} - \frac{1}{4}\right]$$

Tutorial problem (Solved)

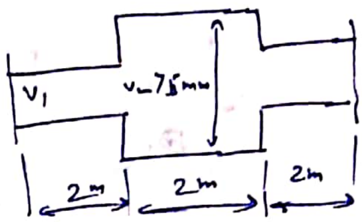
Qu A pipe of 50 mm dia. is 6 m long and the velocity of flow of water in the pipe is 2.4 m/s. What loss of head and the corresponding power would be saved if the central 2 m length of pipe was replaced by 75 mm dia. pipe, the change of section being sudden? Take f = 0.04 for pipe of both diameters.

Solution

Case I



Case II



Case I:

V = 2.4 m/s

Frictional head loss in central 2 m length

$$h_f = \frac{.04 \times 2 \times (2.4)^2}{2 \times 9.81 \times (50 \times 10^{-3})} = 0.47 \text{ m}$$

$A_1V_1 = A_2V_2$

$$V_2 = \frac{\pi/4 \times (50)^2 \times 2.4}{\pi/4 \times (75)^2} = \left(\frac{2}{3}\right)^2 \times 2.4 = 1.067 \text{ m/s}$$

Case II:

Total loss of head Now

loss due to

1) Sudden enlargement

$$h_e = \frac{(V_1 - V_2)^2}{2g} = \frac{(2.4 - 1.067)^2}{2g} = 0.090 \text{ m}$$

2) h_f = friction loss

$$= \frac{.04 \times 2 \times (1.067)^2}{2g \times .075} = .062 \text{ m}$$

3) loss due to sudden contraction h_c

$$= 0.5\frac{V_2^2}{2g} = \frac{0.5(2.4)^2}{2g} = 0.147 \text{ m}$$

Total head loss = 0.147 m + 0.062 m + 0.090 m = 0.299 m

Head Saved = 0.47 − 0.299 = 0.171 m

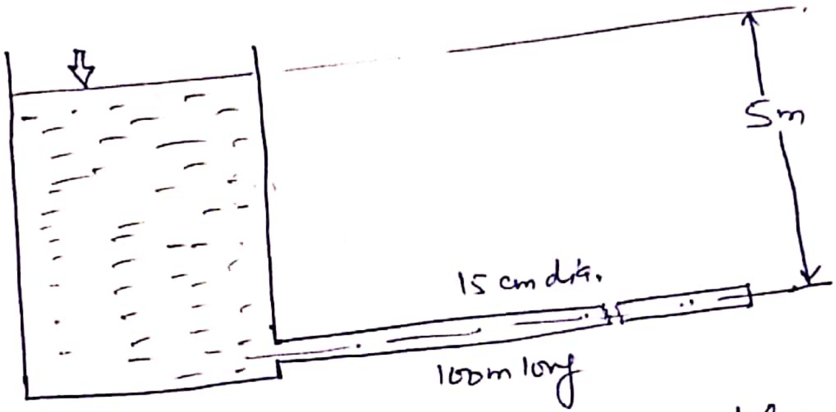
$\boxed{Q = A \times V = \frac{\pi}{4} \times (.05)^2 \times 2.4 = 4.71 \times 10^{-3} \text{ m}^3/\text{s}}$

Power Saved = $\rho g Q h$
= $1000 \times 4.71 \times 10^{-3} \times 9.81 \times 0.171$
= 7.90 Watts.

Tutorial Problem (Solved)

Water is discharged from a tank maintained at a constant head of 5m above the exit of a straight pipe 100m long, 15 cm diameter. Estimate the rate of flow if friction factor for the pipe is given as 0.01.

Solution



This is pipe discharging freely into atmosphere hence head is lost in overcoming friction, entrance loss and exit loss.

$$h_f = \frac{flv^2}{2gd}, \quad h_{entrance} = \frac{0.5v^2}{2g}, \quad h_{exit} = \frac{v^2}{2g}.$$

$$5 = \frac{v^2}{2g}\left[\frac{fl}{d} + 0.5 + 1\right]$$

$$\Rightarrow v = 3.466 \text{ m/s}$$

$$Q = A \times V$$

$$= \frac{\pi}{4} \times (.15)^2 \times 3.466$$

$$= \underline{\underline{0.0612 \text{ m}^3/\text{s}}}$$

Tutorial problem (Solved)

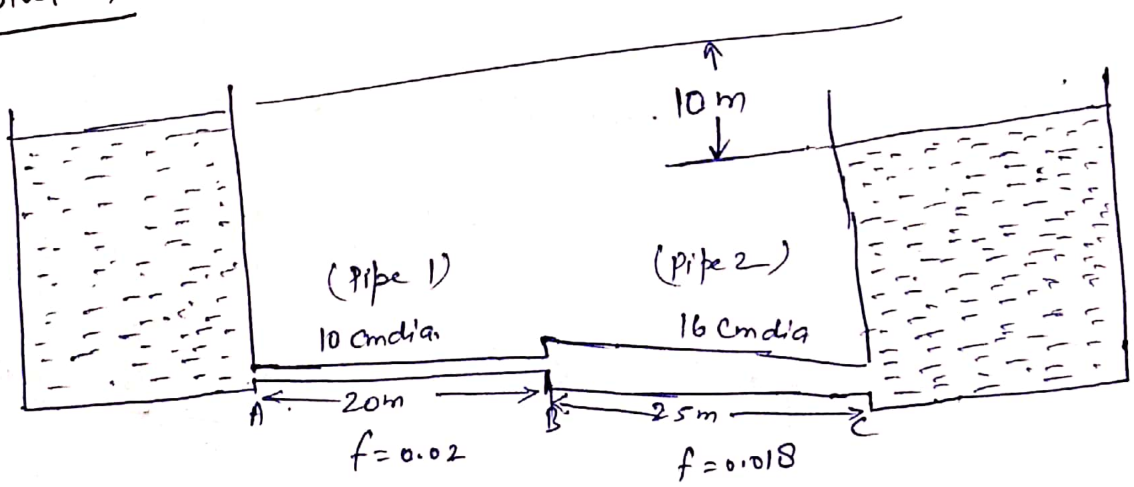
Two reservoirs with a diffrence in water surface elevation of 10m are connected by a pipeline ABC which consists of two pipes AB and BC joined in series. Pipe AB is 10 cm in diameter, 20m long and has a value of $f=0.02$. Pipe BC is of 16 cm diameter, 25 m long and has an $f=0.018$. The junction with the reservoirs and between the pipes are abrupt.

(i) Calculate the discharge.

(ii) What difference in reservoir elevations is necessary to have a discharge of 15 L/s.

Include all minor losses.

Solution



$$\text{Entrance loss} = \frac{0.5 V_1^2}{2g}$$

$$= 0.5 \times \frac{V_1^2}{2g}.$$

$$\text{Loss due to Sudden expansion} = h_e = \frac{(V_1 - V_2)^2}{2g} = \frac{V_1^2}{2g}\left(1 - \frac{V_2}{V_1}\right)^2$$

(23)

$Q = A_1 V_1 = A_2 V_2$

$= \frac{\pi}{4} d_1^2 V_1 = \frac{\pi}{4} d_2^2 V_2$

$\frac{V_2}{V_1} = \left(\frac{d_1}{d_2}\right)^2 = \left(\frac{10}{16}\right)^2 = (.625)^2 = .390625$

$h_e = \frac{V_1^2}{2g}\left[1 - .390625\right]^2 = .37134 \frac{V_1^2}{2g}$

loss at exit $= \frac{V_2^2}{2g} = (.390625)^2 \frac{V_1^2}{2g}$

$= 0.1526 \frac{V_1^2}{2g}$

friction loss in pipe 1 $= h_{f_1} = \frac{f_1 l_1 V_1^2}{2g d_1} = \frac{.02 \times 20 \times V_1^2}{2g \times .10} = 4 \frac{V_1^2}{2g}$

friction loss in pipe 2 $h_{f_2} = \frac{f_2 l_2 V_2^2}{2g d_2} = \frac{.018 \times 25 \times (.390625)^2 V_1^2}{2g \times .16}$

$= .429 \frac{V_1^2}{2g}$

Total head loss $= \frac{V_1^2}{2g}\left[0.5 + .37134 + .1526 + 4 + .429\right] = 5.45294 \frac{V_1^2}{2g}$

(i) head loss $= 10\text{m} = 5.45294 \frac{V_1^2}{2g} \Rightarrow V_1 = 6 \text{ m/s}$

$V_2 = 2.34 \text{ m/s}$

$Q = \frac{\pi}{4} d_1^2 V_1 = \frac{\pi}{4} \times (.10)^2 \times 6 = .0471 \text{ m}^3/\text{s}$

$= 47.1 \text{ L/s}$ ANS

(ii) for $Q = 15 \text{ L/s}$

$= .015 \text{ m}^3/\text{s}$, $V_1 = \frac{.015 \times 4}{\pi \times (.1)^2} = 1.91 \text{ m/s}$

$H_L = 5.45294 \frac{V_1^2}{2g}$

$\Rightarrow H_L = 1.013 \text{ m}$ ANS

Tutorial Problem (Solved)

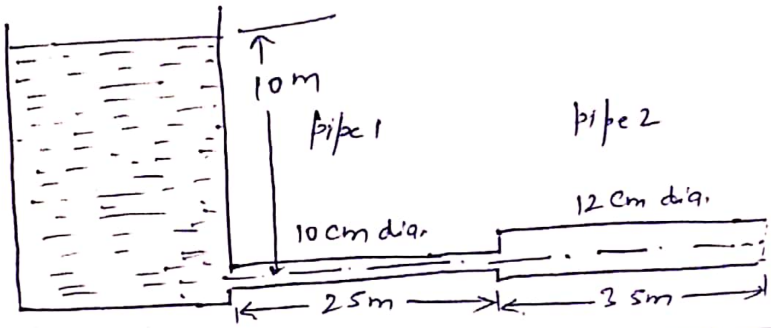
A reservoir discharges its liquid content through a horizontal pipline into the atmosphere. The pipe line consists of two pipes: one of 10 cm diameter and 25 m long and another of 12 cm diameter and 35 long, connected in series. The friction factor $f = 0.02$ for both pipes. The water level in the tank is 10 m above the centreline of pipe at entrance. Consider all minor losses,

(i) Calculate the discharge when 10 cm diameter pipe is joined to tank.

(ii) Calculate the discharge when 12 cm diameter pipe is joined to tank.

assume ($C_c = 0.70$)

Solution (i)



$Q = A_1 V_1 = A_2 V_2$

$V_2 = \frac{A_1}{A_2} V_1 = \left(\frac{d_1}{d_2}\right)^2 V_1$

$V_2 = \left(\frac{10}{12}\right)^2 V_1 = 0.6944 V_1$

entrance loss $= \frac{0.5 V_1^2}{2g}$

enlargement loss $= \frac{(V_1 - V_2)^2}{2g} = \frac{V_1^2}{2g}\left(1 - \frac{V_2}{V_1}\right)^2$

$= \frac{V_1^2}{2g}(1 - .6944)^2$

$= 0.09336 \frac{V_1^2}{2g}$

exit loss $= \frac{V_2^2}{2g} = (.6944)^2 \frac{V_1^2}{2g} = .4822 \frac{V_1^2}{2g}$

friction loss in 10 cm pipe $= h_{f_1} = \frac{f_1 l_1 V_1^2}{2g d_1} = \frac{.02 \times 25 \times V_1^2}{.1 \times 2g}$

$= 5 \frac{V_1^2}{2g}$

friction loss in 12 cm pipe $= h_{f_2} = \frac{f_2 l_2 V_2^2}{2g d_2} = \frac{.02 \times 35 \times (.6944)^2}{.12} \frac{V_1^2}{2g}$

$= 2.81 \frac{V_1^2}{2g}$

Total head loss $h_L = \frac{V_1^2}{2g}\left[0.5 + .09336 + .4822 + 5 + 2.81\right] = 8.8856 \frac{V_1^2}{2g}$

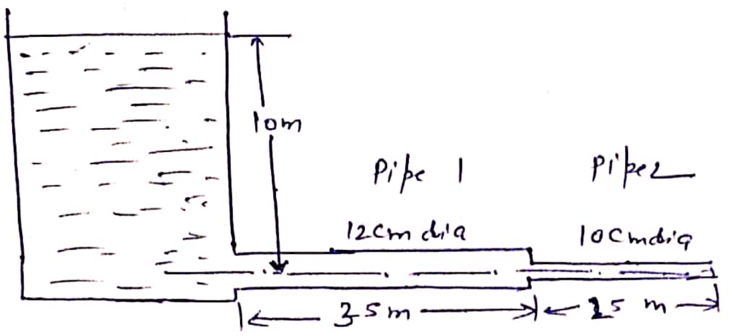
(25)

$$0.8856 \frac{V_1^2}{2g} = 10 \Rightarrow V_1 = 4.70 \text{ m/s}$$

$$Q = \frac{\pi}{4} d_1^2 \times V_1 = \frac{\pi}{4} \times (.1)^2 \times (4.7) = .0.0369 \text{ m}^3/\text{s}$$

$$= \underline{\underline{36.9 \ \ell/s}}$$

(ii)



entrance loss $= 0.5 \frac{V_1^2}{2g}$.

Contraction loss $= \frac{V_2^2}{2g}\left(\frac{1}{C_c}-1\right)^2 = \frac{V_2^2}{2g}\left(\frac{1}{.7}-1\right) = .1837 \frac{V_2^2}{2g} = .3809 \frac{V_1^2}{2g}$

$\frac{\pi}{4} d_1^2 V_1 = \pi/_4 d_2^2 V_2 \Rightarrow V_2 = \left(\frac{d_1}{d_2}\right)^2 V_1 = 1.44 V_1$

exit loss $= \frac{V_2^2}{2g} = (1.44)^2 \frac{V_1^2}{2g} = 2.0736 \frac{V_1^2}{2g}$

h_{f_1} = head loss due to friction in 12 cm pipe

$$= \frac{f_1 l_1 V_1^2}{2g d_1} = \frac{.02 \times 35 \times V_1^2}{.12 \times 2g} = 5.833 \frac{V_1^2}{2g}.$$

h_{f_2} = head loss due to friction in 10 cm pipe

$$= \frac{f_2 l_2}{d_2} \frac{V_2^2}{2g} = \frac{.02 \times 25 \times (1.44)^2}{.1} \frac{V_1^2}{2g} = 10.368 \frac{V_1^2}{2g}$$

total head loss $= \frac{V_1^2}{2g}\left[0.5 + .3809 + 2.0736 + 5.833 + 10.368\right]$

$$10 = 19.1555 \frac{V_1^2}{2g}.$$

$$V_1 = 3.2 \text{ m/s}$$

$$Q = \pi/_4 d_1^2 \times V_1$$

$$= \pi/_4 \times (.12)^2 \times 3.2 = 0.0362 \text{ m}^3/\text{s}$$

$$= \underline{\underline{36.2 \ \ell/s}}.$$

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Laminar flow

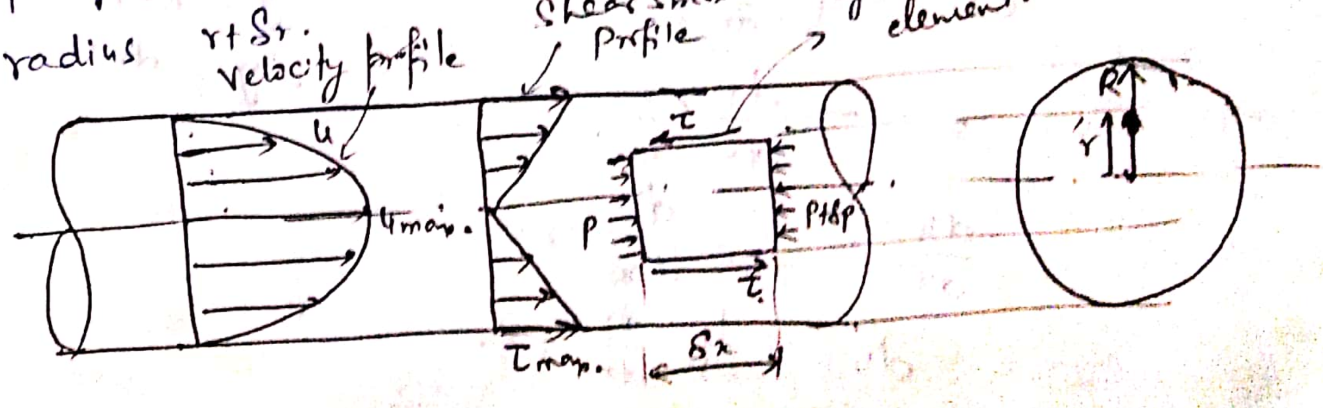
- Fluid particles move along straight parallel paths in layers or laminae.
- Laminar flow occurs at low velocity so that forces due to viscosity predominate over inertial forces.
- Fluid rise over each other giving rise to shear stresses being maximum at boundary. Shear stresses so produced ~~developed~~ resistance to flow.
- In order to overcome resistance pressure drops from point to point along flow direction.
- Relationship between shear and pressure gradient in laminar flow

$$\frac{\partial \tau}{\partial y} = \frac{\partial P}{\partial x}$$

Steady laminar flow through a circular pipe:

Hagen poiseullie's law

Consider steady laminar flow of an incompressible real fluid through a horizontal circular pipe. Since the flow proceeds in circular laminae, we take equilibrium of the flow contained in the lamina of of cylinder of radius r radius r+δr.



②

For equilibrium net pressure force on cylindrical element due to pressure P & P+δP must be equal to shear force on its periphery due to presence of cylindrical lamina over it.

$$P \times \pi r^2 - (P+\delta P)\pi r^2 = \tau \times 2\pi r \times \delta x$$

$$\tau = -\frac{dp}{dx}\frac{r}{2} \quad \text{———} (i)$$

If laminar flow in a pipe is fully developed, the velocity profile do not change in longitudinal direction, the velocity u varies only with r and pressure p remains constant over the cross section.

Hence $\frac{dp}{dx}$ remains constant.

$\boxed{\frac{dp}{dx} = \text{Constant}} \Rightarrow \boxed{\tau \propto r}$ Shear stress varies linearly with radius.

τ_{max} is at $r = R$

$$\tau_{max} = -\frac{dp}{dx}\frac{R}{2}$$

Velocity distribution

For laminar flow

$$\tau = \mu \frac{du}{dy}$$

→ Since R is taken from centre
→ y is measured from wall
$y = R - r$
$dy = -dr$

$$\tau = \mu\left(-\frac{du}{dr}\right)$$

$$= -\frac{dp}{dx}\frac{r}{2}$$

$$\frac{du}{dr} = \frac{1}{2\mu}\frac{dp}{dx} r$$

$$du = \frac{1}{2\mu}\frac{dp}{dx} r\,dr$$

$$u = \frac{1}{2\mu}\frac{dp}{dx}\frac{r^2}{2} + c$$

B.C. at $\boxed{r = R, \quad u = 0}$

$$\Rightarrow c = -\frac{1}{4\mu}\frac{dp}{dx}(R^2)$$

$$u = \frac{1}{4\mu}\frac{dp}{dx}(r^2 - R^2)$$

$$\boxed{u = -\frac{1}{4\mu}\frac{dp}{dx}(R^2 - r^2)}$$ ← Parabolic velocity distribution.

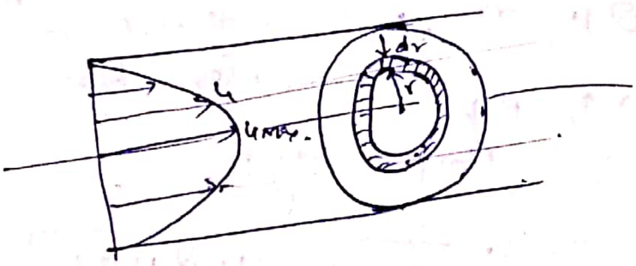
B.C at $r = 0$, $u = u_{max}$.

$$u_{max} = -\frac{1}{4\mu}\frac{dp}{dx}R^2$$

$$\boxed{u = -\frac{1}{4\mu}\frac{dp}{dx}R^2\left[1 - \frac{r^2}{R^2}\right]}$$

$$\boxed{u = u_{max}\left[1 - \frac{r^2}{R^2}\right]}$$

Ratio of max^n velocity to average velocity.



Total discharge (Q) passing through pipe can be obtained by integrating small discharge passing through elemental ring of thickness dr.

$$dq = [(2\pi r)\,dr] \times u$$

$$q = \pi R^2 u_{mean}.$$

$$\pi R^2 u_{mean} = \int_0^R u(2\pi r\,dr)$$

$$u_{mean} = \frac{1}{\pi R^2}\int_0^R u\,dA$$

$$\left[\begin{array}{l} A = \pi R^2 \\ dA = 2\pi r\,dr \end{array}\right]$$

$$\boxed{u_{mean} = \frac{1}{A}\int_0^R u\,dA}$$

$$u_{mean} = \frac{1}{\pi R^2}\int_0^R \underbrace{u_{max}\left(1-\frac{r^2}{R^2}\right)} 2\pi r\,dr$$

$$\boxed{u = u_{max}\left(1-\frac{r^2}{R^2}\right)}$$

$$u_{mean} = \frac{u_{max}}{\cancel{\pi} R^2}\, 2\cancel{\pi}\int_0^R \left(r - \frac{r^3}{R^2}\right)dr$$

$$= \frac{2u_{max}}{R^2}\left[\frac{r^2}{2} - \frac{r^4}{4R^2}\right]_0^R = \frac{2u_{max}}{\cancel{R^2}}\cdot\frac{\cancel{R^2}}{4}$$

$$u_{mean} = \frac{u_{max}}{2} \Rightarrow \boxed{u_{max} = 2u_{mean}}$$

<u>Discharge</u>

$$q = A \times V$$

$$= \frac{\pi}{4}D^2 \times u_{mean} = \frac{\pi}{4}D^2 \times \frac{u_{max}}{2}$$

$$= \frac{\pi}{4}D^2 \times \frac{-\frac{1}{4\mu}\frac{dp}{dx}R^2}{2} \quad D^2/4$$

$$\boxed{q = -\frac{\pi D^4}{128\mu}\frac{dp}{dx}}$$

Hagen poiseullie's equation for laminar flow through a circular pipe.

<u>Note:</u>
$$\frac{dp}{dx} = -\frac{128\mu q}{\pi D^4}$$

as $x\uparrow$, $p\downarrow$ therefore $-ve$.

$$\frac{P_1 - P_2}{L} = \frac{128\mu q}{\pi D^4} \Rightarrow \boxed{P_1 - P_2 = \frac{128\mu q L}{\pi D^4}}$$

$$\boxed{P_1 - P_2 = \frac{32\mu V L}{D^2}}$$

$$h_f = \frac{P_1 - P_2}{\rho g} = \frac{128\mu q L}{\pi D^4 \cdot \rho g}$$

$$= \frac{128\mu \times \left(\frac{\pi}{4}D^2 \times V\right) \times L}{\pi D^4 \rho g}$$

$$\boxed{h_f = \frac{32\mu V L}{\rho g D^2}}$$

Hagen's poiseulles formular for laminar flow.

$h_f \propto V$
$h_f \propto \frac{1}{D^2}$ for lamina flow.

Power to maintain the flow

$$\boxed{\text{Power} = \left[(P_1 - P_2) \times \pi R^2 \right] \times V}$$

Total drag force on pipe

$$\boxed{D = \tau_{max} \times 2\pi R \times l}$$

Tutorial problem (Solved)

An oil having viscosity of 0.0098 kgf-second/m^2 and a specific gravity of 1.59 flows through a horizontal pipe of 5 cm diameter with a pressure drop of 0.06 kgf/cm^2 per meter length of pipe.
Determine (i) flow rate in kg/min.
(ii) Shear stress at the pipe wall.
(iii) The total drag force for 100 m length of pipe, and
(iv) The power required for the 100 m length of pipe to maintain the flow.

Solution:

$$\mu = .0098 \text{ kgf second}/m^2 = .0098 \times 9.8 \ \frac{N\text{-}s}{m^2}$$
$$= .096 \ \frac{Nm}{m^2}$$

pressure drop = .06 kgf/cm^2 per m pipe length

$$\Delta P = \frac{.06 \times 9.8}{10^{-4}} \ N/m^2 \text{ per m pipe length}$$
$$= 5880$$

$$\frac{dp}{dx} = -5880 \ N/m^2 \text{ per meter pipe length}$$

$$Q = \frac{-\pi D^4}{128 \mu} \frac{dp}{dx}$$

(i)
$$Q = \frac{-3.14 \times (.05)^4}{128 \times .096} \times -5880$$
$$= .00939 \ m^3/s$$

$$\rho = 1.59 \times 1000 = 1590 \ kg/m^3$$

$$\dot{m} = \rho Q = 1590 \times .00939$$
$$= 14.93 \ kg/s$$
$$= 895.80 \ kg/min.$$

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(ii) Shear Stress at pipe wall = τ_{max}.

$$\tau_{max} = -\frac{dp}{dx}\frac{R}{2}$$

$$= -(-5880) \times \frac{.05/2}{2} = \underline{\underline{73.5\ N/m^2}}$$

(iii)

Total drag force on 100 m length of pipe

$$= \tau_{max} \times 2\pi R \times l$$

$$= 73.5 \times 2 \times 3.14 \times \frac{.05}{2} \times 100$$

$$= 1153.95\ N$$

$$= \underline{\underline{1.154}}\ kN$$

(iv) Power required to maintain the flow

$$= (P_1 - P_2)\ \pi R^2\ U_{mean}$$

$$U_{mean} = \frac{U_{max.}}{2}$$

$$u_{max.} = -\frac{1}{4\mu}\frac{dp}{dx}R^2$$

$$= -\frac{1}{4 \times .096} \times (-5880) \times (.025)^2$$

$$= 9.57\ m/s$$

$$U_{mean} = 4.785\ m/s$$

$$\text{Power} = 5880 \times 3.14 \times (.025)^2 \times 4.785$$

$$= 55.216\ W/\text{metre length}$$

for 100 m length $= 5521\ W$

$$= \underline{\underline{5.521}}\ kW.$$

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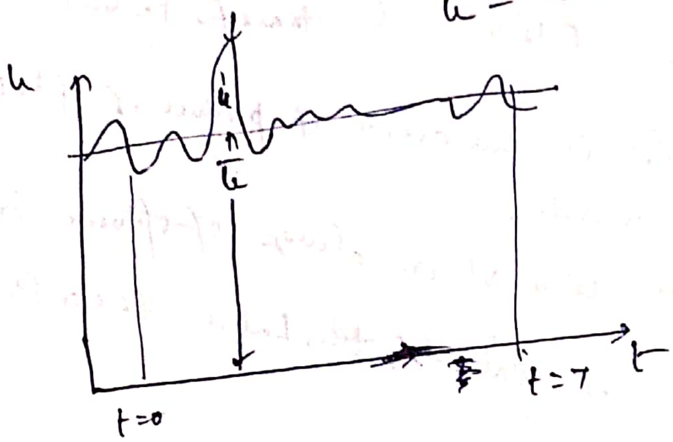
Turbulent flow

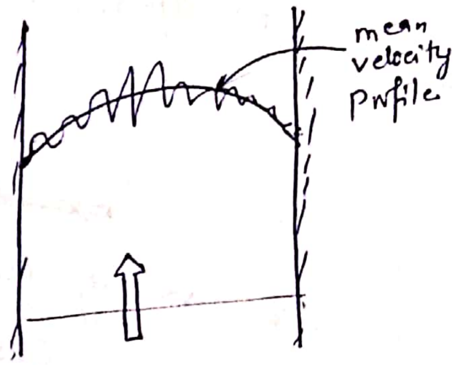
- Turbulence is random phenomenon where various quantities related to flow vary randomly with time and space.
- The theory of turbulence is based upon statistical considerations
- Instantaneous velocity u in a turbulent flow can be considered as sum of time-averaged velocity $\bar{u}$ and fluctuating component u'.

u' may be +ve or -ve

$$u = \bar{u} + u'$$



Temporal fluctuations.



Instantaneous pattern.

Similarly instantaneous v-component of velocity, pressure and density are related as

$$v = \bar{v} + v'$$
$$p = \bar{p} + p'$$
$$\rho = \bar{\rho} + \rho'$$

By definition time average velocity is given as

$$\bar{u} = \frac{1}{T}\int_0^T u\,dt$$

$$\bar{u'} = \frac{1}{T}\int_0^T u'\,dt = 0$$ (i.e. time averaged fluctuations must vanish)

Intensity of turbulence is given as I, It is r.m.s. value of fluctuations

$$I = \sqrt{\overline{u'^2}}$$

% of turbulence in the direction of free stream $\frac{I}{U} \times 100$

Turbulent Shear Stress

The mechanism of turbulence is responsible for large scale momentum exchange which results in apparent shear stress (τ_t)

$$\tau_t = -\rho\overline{u'v'} \quad (\text{mean rate of momentum transfer per unit area})$$

~~instantaneous~~ $\overline{u'v'}$ is time average of product of fluctuating velocity component u' and v'.

experiment shows that u' & v' are always of opposite sign

hence $-\rho\overline{u'v'}$ is called turbulent shear stress or Reynolds stress.

Boussinesq eddy viscosity

In analogy with coefficient of viscosity for laminar flow Boussinesq defined mixing coefficient also known as turbulent viscosity

$$\tau_t = -\rho\overline{u'v'} = (\mu_t)\frac{d\bar{u}}{dy}$$

μ_t → eddy viscosity or turbulent viscosity (much greater than laminar viscosity)

Total viscosity (effective viscosity)

$$\boxed{\mu_{eff} = \mu + \mu_t}$$

Prandtl mixing length theory

Prandtl mixing length theory is based on concept of momentum exchange between fluids.

Prandtl hypothesised that each fluctuating component is proportional to average axial velocity gradient and constant of proportionality is called prandtl mixing length such that

$$u' = l_1 \frac{d\bar{u}}{dy}$$

$$v' = l_2 \frac{d\bar{u}}{dy}. \qquad l_1 \simeq l_2 \simeq l$$

$$\overline{u'v'} = -l^2 \left(\frac{d\bar{u}}{dy}\right)^2$$

Equation does not show a sign change of $\frac{d\bar{u}}{dy}$ Hence we write

$$\boxed{\tau_t = -\rho\overline{u'v'} = \rho l^2 \left|\frac{d\bar{u}}{dy}\right| \left(\frac{d\bar{u}}{dy}\right)}$$

This equation is known as prandtl mixing length hypothesis.

Prandtl suggested that for pipe flow length l must be proportional to distance y from the wall as the mixing movement of u' and v' must reduce to zero at solid wall

$$\boxed{l = ky}$$

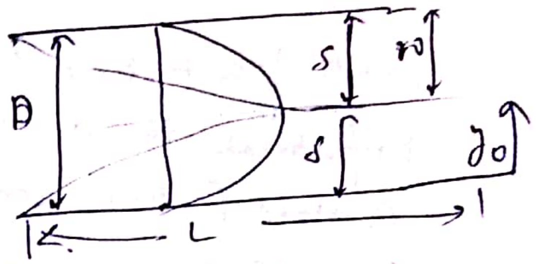
K = proportionality constant called universal constant and is determined from experiments.

(4)

Shear velocity or friction velocity

For a fully developed pipe flow pressure force along the flow is balanced by shear force at the pipe wall.

δ = boundary layer thickness
$= r_0 = D/2 =$ Constant



Shear stress varies linearly with distance from pipe wall as

$$\tau = \tau_0 (1 - y_0/r_0)$$

Wall at $y_0 = 0$, $\tau = \tau_0$
Centre at $y_0 = r_0$, $\tau = 0$

Pressure force = Shear force

$$(P_1 - P_2)\frac{\pi}{4} D^2 = \tau_0 \times \pi D L$$

$$\tau_0 = \frac{P_1 - P_2}{\rho g} \rho g \frac{D^2}{4 D L}$$

$$= h_f \cdot \rho g \cdot \frac{D}{4L}$$

$$= \frac{f L V^2}{2 g D} \cdot \rho g \frac{D}{4L}$$

$$\tau_0 = \frac{f V^2 \rho}{8}$$

$$\frac{\tau_0}{\rho} = V^2 \frac{f}{8} \Rightarrow \sqrt{\frac{\tau_0}{\rho}} = V\sqrt{f/8}$$

$$\boxed{u_* = V\sqrt{f/8} = \sqrt{\frac{\tau_0}{\rho}}}$$

u_* is called shear velocity or friction velocity.